# Einstein-Residual Closure Standard: Invariant Dark-Sector Audits, Normal Forms, and Topology-Likelihood Boundaries

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A dark-sector model should not be evaluated before its geometric residual, visible-sector convention, homogeneous normal form, microscopic realization, amplitude selector, perturbative transfer map, and likelihood benchmark have been separately declared. We formulate this requirement as an Einstein-residual closure standard: a convention-explicit audit protocol rather than a new local law of nature. For a fixed spacetime metric, a declared Einstein-side convention, and a specified visible-sector subtraction, the Einstein residual

$$T^{\mathcal{D}}_{\mu\nu} = \frac{c^4}{8\pi G} G_{\mu\nu} - T^{\rm vis}_{\mu\nu}$$

is a covariantly fixed reconstruction target rather than a source assumption. In the homogeneous locally-FLRW sector the residual admits the lossless atlas

$$\mathcal{A}_q = -p_{\mathcal{D}}, \qquad \mathcal{A}_{\rm w} = a^3(\epsilon_{\mathcal{D}} + p_{\mathcal{D}}),$$

and is exactly minimal dust plus fixed vacuum if and only if

$$\frac{{\rm d}\mathcal{A}_q}{{\rm d}\ln a} = 0, \qquad \frac{{\rm d}\mathcal{A}_{\rm w}}{{\rm d}\ln a} = 0, \qquad \mathcal{A}_q, \mathcal{A}_{\rm w} \geq 0.$$

This theorem is algebraic, not an abundance-origin mechanism. Its minimal background realization is a conserved Brown current with linear energy law plus a fixed four-form vacuum branch. Brown-current dynamics closes the dust equation of state but not the dust abundance; the four-form closes $w = -1$ but not the vacuum amplitude. Compact causal geometry supplies conformal null-reach invariants, and compact topology enters through quotient projectors, spectra, observable covariance, and observer priors; neither selects $\Omega_c$, $\Omega_\Lambda$, or $\Lambda_{\rm eff}$ by itself. We make the standard operational by introducing an admissibility vector, hierarchy no-go rules, closure-defect and likelihood ledgers, and a minimal benchmark interface. The resulting principle is: no dark-sector claim without residual reconstruction, closure-defect test, microscopic ledger, amplitude ledger, transfer ledger, and matched benchmark likelihood.

## I. CLAIM BOUNDARY

This paper gives a conditional homogeneous stress-tensor classification. The classification is algebraic rather than dynamical: it does not derive the observed dark-sector abundances, does not select a flux branch, and does not claim detection of compact topology. No abundance-origin claim is made. Once the metric $g_{\mu\nu}$, the Einstein-side convention, and the visible-sector subtraction $T^{\rm vis}_{\mu\nu}$ are fixed, the Einstein residual is fixed. For that same convention and visible subtraction, if this residual is homogeneous and locally FLRW, then it has the minimal dust-like plus fixed-vacuum stress-tensor form if and only if

$$-p_{\mathcal{D}} = \text{constant} \geq 0, \qquad a^3(\epsilon_{\mathcal{D}} + p_{\mathcal{D}}) = \text{constant} \geq 0. \tag{1}$$

Microscopic current and four-form realizations may impose further amplitude selection rules; the closure theorem itself does not supply them. The corresponding stress tensor is locally indistinguishable from cold dark matter plus $\Lambda$ with the same amplitudes and curvature. Any distinction must therefore come from compact spectral projection, quotient covariance, perturbative null conditions, or an additional amplitude-selection rule.

The paper uses a ledger hierarchy. Level I is residual reconstruction from $(g_{\mu\nu}, T^{\rm vis}_{\mu\nu})$. Level II is the homogeneous normal form in $(\mathcal{A}_q, \mathcal{A}_{\rm w})$. Level III is Brown-current realization of the dust equation of state. Level IV is fixed four-form realization of the vacuum equation of state. Level V is compact causal reach, which constrains null accessibility but adds no new homogeneous source. Level VI is perturbative source behavior, such as dust sound speed and anisotropic-stress null conditions. Level VII is a topology-likelihood claim, where the quotient projector has been propagated into a masked, noisy observable covariance. This manuscript supplies the Level-I/II normal-form theorem and disciplined completion ledgers for Levels III–VII; it does not perform a completed topology likelihood or amplitude selection.

The local residual theorem applies to any FLRW curvature class once the homogeneous metric, the visible subtraction, and the Einstein-side convention are declared.

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The compact topology ledger used below is narrower: it is specialized to compact orientable spherical space forms $S^3_A/\Gamma$, except where Euclidean compact-topology papers are cited explicitly as covariance-level methodological analogues.

The paper is complete only at this declared level: homogeneous residual closure, compact-background bookkeeping, perturbative null conditions, and amplitude-selection boundaries. It is not a compact Boltzmann solver, a likelihood analysis, or a microscopic dark-sector completion.

## II. EINSTEIN-RESIDUAL CLOSURE STANDARD

The preceding boundary can be elevated from a single model statement to a reproducible standard. The standard is not a new local law of nature. It is a claim-control protocol for dark-sector models: each model must identify which part of the argument is reconstruction, which part is microscopic realization, which part is amplitude selection, and which part is observational likelihood.

**Law 1** (Primitive residual law)**.** *Given $(g_{\mu\nu}, T^{\rm vis}_{\mu\nu})$ and an Einstein-side convention, the only primitive dark-sector reconstruction target is*

$$T^{\mathcal{D}}_{\mu\nu}[g,V] = \frac{c^4}{8\pi G}G_{\mu\nu}[g] - T^{\rm vis}_{\mu\nu}[V]. \tag{2}$$

*It is a residual tensor, not automatically a particle species, vacuum branch, topological sector, or modified-gravity mechanism.*

**Law 2** (Subtraction covariance)**.** *If the declared visible subtraction is changed by a homogeneous perfect-fluid piece $X_{\mu\nu} = \epsilon_X U_\mu U_\nu/c^2 + p_X h_{\mu\nu}$, then*

$$T^{\mathcal{D}}_{\mu\nu} \mapsto T^{\mathcal{D}}_{\mu\nu} - X_{\mu\nu}, \tag{3}$$

*and the homogeneous atlas transforms affinely as*

$$\mathcal{A}_q \mapsto \mathcal{A}_q + p_X, \qquad \mathcal{A}_{\rm w} \mapsto \mathcal{A}_{\rm w} - a^3(\epsilon_X + p_X). \tag{4}$$

*Consequently, no residual-closure claim is invariant until the visible subtraction and Einstein-side convention have been declared.*

**Law 3** (FLRW atlas completeness)**.** *In the homogeneous locally-FLRW perfect-fluid class,*

$$(\epsilon_{\mathcal{D}}, p_{\mathcal{D}}) \longleftrightarrow (\mathcal{A}_q, \mathcal{A}_{\rm w}) \tag{5}$$

*with inverse*

$$p_{\mathcal{D}} = -\mathcal{A}_q, \qquad \epsilon_{\mathcal{D}} = a^{-3}\mathcal{A}_{\rm w} + \mathcal{A}_q. \tag{6}$$

*Thus any further distinction between two models with the same atlas is not a homogeneous-background distinction; it must arise from microscopic realization, amplitude selection, perturbations, topology, or likelihood.*

**Law 4** (Minimal closure law)**.** *The homogeneous residual is exactly minimal dust plus fixed vacuum on a connected scale-factor interval $I$ if and only if*

$$\frac{\mathrm{d}\mathcal{A}_q}{\mathrm{d}\ln a} = 0, \qquad \frac{\mathrm{d}\mathcal{A}_{\rm w}}{\mathrm{d}\ln a} = 0, \qquad \mathcal{A}_q, \mathcal{A}_{\rm w} \geq 0. \tag{7}$$

*In that case*

$$T^{\mathcal{D}}_{\mu\nu} = \epsilon_{{\rm w},0}a^{-3}\frac{U_\mu U_\nu}{c^2} - \epsilon_q g_{\mu\nu}. \tag{8}$$

*This is an iff statement for the homogeneous stress tensor; it is not an amplitude-origin theorem.*

**Law 5** (Closure-defect invariant)**.** *For any reference density $\epsilon_* > 0$, define*

$$\mathfrak{C}_I[\mathcal{D};\epsilon_*] := \sup_{a\in I}\frac{\left(\frac{\mathrm{d}\mathcal{A}_q}{\mathrm{d}\ln a}\right)^2 + \left(a^{-3}\frac{\mathrm{d}\mathcal{A}_{\rm w}}{\mathrm{d}\ln a}\right)^2}{\epsilon_*^2}. \tag{9}$$

*Then $\mathfrak{C}_I = 0$ if and only if the adopted residual is exactly minimal dust plus fixed vacuum on $I$. The value depends on the chosen norm and reference density, but the zero set does not.*

**Law 6** (Brown-current dust law)**.** *For a Brown-current realization with $N^\mu = nu^\mu$, $\nabla_\mu N^\mu = 0$, and energy law $F(n)$,*

$$\epsilon_J = F(n), \qquad p_J = nF'(n) - F(n). \tag{10}$$

*The pressureless condition is equivalent to*

$$nF'(n) = F(n), \tag{11}$$

*whose regular nonzero local solution is*

$$F(n) = \mu_{\rm w}c^2 n. \tag{12}$$

*Thus dust behavior is linear current energy, not topology.*

**Law 7** (Current-amplitude no-go)**.** *A topological degree $Q \in \mathbb{Z}$ labels a sector. It becomes a Brown-current number only after the additional branch rule*

$$N_J = |Q| \tag{13}$$

*has been imposed. Even then,*

$$\Omega_{\rm w} = \frac{\mu_{\rm w}c^2|Q|}{\mathrm{Vol}(\Sigma_{A_0})\epsilon_{{\rm crit},0}} \tag{14}$$

*requires the non-topological scale $\mu_{\rm w}$. Therefore $(Q,\Gamma)$ by themselves do not determine $\Omega_{\rm w}$.*

**Law 8** (Four-form vacuum law)**.** *A four-form branch in four spacetime dimensions obeys, on a fixed connected branch,*

$$F_{\mu\nu\rho\sigma} = f\varepsilon_{\mu\nu\rho\sigma}, \qquad \mathrm{d}f = 0, \tag{15}$$

*and therefore has vacuum stress tensor*

$$T^q_{\mu\nu} = -\epsilon_q g_{\mu\nu}. \tag{16}$$

*It closes the equation of state $w = -1$, not the observed amplitude $\epsilon_q$.*

**Law 9** (Flux-lattice separation)**.** *The continuous fixed-vacuum statement*

$$\epsilon_q = \frac{1}{2}\widehat{q}^2, \qquad \widehat{q} \in \mathbb{R}, \tag{17}$$

*is distinct from the added flux-lattice rule*

$$\widehat{q} = N_F e_F, \qquad N_F \in \mathbb{Z}. \tag{18}$$

*Flux quantization requires a normalization or membrane sector; it is not a consequence of compact spatial topology alone.*

**Law 10** (Topological decoupling)**.** *There is no dimensionally complete topology-only map*

$$(\Gamma, Q, N_F) \not\Rightarrow (\Omega_{\rm w}, \Omega_q, \Lambda_{\rm eff}). \tag{19}$$

*Any amplitude selector must factor through additional non-topological data, for example*

$$\mathfrak{S}_{\rm amp} : (\Gamma, Q, N_F; \mu_{\rm w}, e_F, A_0, H_0, \mathcal{B}) \longrightarrow (\Omega_{\rm w}, \Omega_q). \tag{20}$$

**Law 11** (Compact spectral projector)**.** *For $\Sigma_\Gamma = S^3/\Gamma$, physical fields are projected cover fields,*

$$\Phi(\gamma x) = \Phi(x), \qquad \gamma \in \Gamma, \tag{21}$$

*with projector*

$$P_\Gamma = \frac{1}{|\Gamma|}\sum_{\gamma\in\Gamma}\gamma^*. \tag{22}$$

*Topology enters through $P_\Gamma$, $\mathrm{Spec}_\Gamma$, $m_\Gamma(n)$, observable covariance, and observer priors, not through a new local homogeneous equation of state.*

**Law 12** (Compact causal accessibility)**.** *For a round simply connected cover,*

$$\Delta\chi(t_1, t_2) = \int_{t_1}^{t_2} \frac{c\,\mathrm{d}t}{a(t)A_0}. \tag{23}$$

*Full antipodal reach on $S^3$ requires $\Delta\chi \geq \pi$. For a quotient, the observer-dependent sufficient reach condition is*

$$\Delta\chi \geq D_\Gamma(x_o), \qquad D_\Gamma(x_o) = \sup_{y\in\Sigma_\Gamma} d_\Gamma(x_o, y). \tag{24}$$

*Compact causality constrains null reach; it does not select energy density.*

**Law 13** (Perturbative degeneracy)**.** *Within a declared data model, the same background amplitudes, visible sector, primordial statistics, and compact transfer functions imply the same observable covariance:*

$$\begin{aligned} &\textit{same } (\epsilon_{\rm w,0}, \epsilon_q), \quad \textit{same visible sector},\\ &\textit{same } \Xi^\Gamma, \quad \textit{same } \Delta_{\ell n}^{X,\Gamma} \Longrightarrow \textit{same } \mathrm{Cov}^{XY,\Gamma}. \end{aligned} \tag{25}$$

*No perturbative observable can distinguish two models after this covariance has been matched.*

**Law 14** (Benchmark likelihood)**.** *A topology claim must compare against a matched benchmark,*

$$\Delta\ln\mathcal{L} = \ln\mathcal{L}_\Gamma - \ln\mathcal{L}_{\rm bench}, \tag{26}$$

*where the benchmark has the same local expansion history, curvature, visible sector, amplitudes, mask, noise, and foreground conventions. Otherwise the comparison confounds topology with background or data-model choices.*

The standard can be summarized as a single admissibility principle:

> A dark-sector claim is admissible only after its Einstein residual, closure atlas, microscopic realization, amplitude selector, perturbative transfer map, and benchmark likelihood have been separately declared and invariantly tested.

## III. UNIVERSAL ADMISSIBILITY INVARIANTS

The closure standard becomes useful beyond this manuscript only if it separates universal bookkeeping from model-specific dynamics. The universal object is not "dark matter," "dark energy," or "topology." It is the ordered declaration

$$\mathfrak{M} = (g_{\mu\nu}, \mathcal{E}, T^{\rm vis}_{\mu\nu}, I, \mathcal{S}_{\rm micro}, \mathcal{S}_{\rm amp}, \mathcal{T}_{\rm pert}, \mathcal{D}, \mathcal{L}_{\rm bench}), \tag{27}$$

where $\mathcal{E}$ denotes the Einstein-side convention, $I$ the scale-factor interval, $\mathcal{S}_{\rm micro}$ the microscopic source ledger, $\mathcal{S}_{\rm amp}$ the amplitude selector, $\mathcal{T}_{\rm pert}$ the perturbative transfer map, $\mathcal{D}$ the declared data vector, and $\mathcal{L}_{\rm bench}$ the matched benchmark likelihood. A model that omits one entry may still be mathematically meaningful, but it is not entitled to the corresponding stronger claim.

**Definition 15** (Admissibility vector)**.** For a declared model $\mathfrak{M}$, define the binary or graded vector

$$\mathbf{A}(\mathfrak{M}) = (A_{\rm R}, A_{\rm H}, A_{\rm J}, A_q, A_{\rm amp}, A_{\rm T}, A_{\rm L}), \tag{28}$$

where $A_{\rm R}$ records residual reconstruction, $A_{\rm H}$ homogeneous closure, $A_{\rm J}$ current or matter-source realization, $A_q$ vacuum realization, $A_{\rm amp}$ amplitude selection, $A_{\rm T}$ perturbative transfer closure, and $A_{\rm L}$ likelihood-grade comparison. The minimal binary convention is

$$A_i \in \{0, 1\}, \tag{29}$$

with $A_i = 1$ only when the corresponding row of the audit table has been explicitly supplied. A numerical implementation may replace the binary entries by calibrated defects or Bayes factors, but the order of the entries is fixed.

**Law 16** (Admissibility monotonicity)**.** *A higher-level claim cannot be inferred from a lower-level entry of* **A**. *In*

*particular,*

$$\begin{aligned} A_{\rm H}=1 &\not\Rightarrow A_{\rm amp}=1,\\ A_{\rm amp}=1 &\not\Rightarrow A_{\rm L}=1,\\ A_{\rm T}=1 &\not\Rightarrow A_{\rm L}=1. \end{aligned} \tag{30}$$

*Thus a background closure theorem, a source EFT, an amplitude selector, and a likelihood claim are logically distinct achievements. This is the basic anti-overclaim invariant of the standard.*

**Law 17** (Hierarchy no-go)**.** *Within the declared FLRW residual standard,*

$$\begin{aligned} \textit{closure atlas} &\not\Rightarrow \textit{microphysics},\\ \textit{Brown current} &\not\Rightarrow \textit{dust abundance},\\ \textit{four-form branch} &\not\Rightarrow \textit{vacuum amplitude},\\ \textit{compact quotient} &\not\Rightarrow \textit{dark-sector amplitudes},\\ \textit{causal reach} &\not\Rightarrow \textit{energy density},\\ \textit{projected modes} &\not\Rightarrow \textit{topology detection}. \end{aligned} \tag{31}$$

*Each implication becomes meaningful only after the missing ledger is supplied: source action, mass or flux scale, branch prior, transfer functions, data vector, and matched likelihood.*

**Theorem 18** (Standard completeness at the declared level)**.** *For homogeneous locally-FLRW residuals with fixed $(g_{\mu\nu}, \mathcal{E}, T^{\rm vis}_{\mu\nu})$, the pair $(\mathcal{A}_q, \mathcal{A}_{\rm w})$, the closure defect $\mathfrak{C}_I$, and the sign conditions $\mathcal{A}_q, \mathcal{A}_{\rm w} \geq 0$ exhaust all background information relevant to whether the residual is minimal dust plus fixed vacuum. Any further question belongs to exactly one of the following ledgers: source realization, amplitude selection, perturbative transfer, compact causal/spectral structure, or likelihood comparison.*

*Proof.* The map $(\epsilon_{\mathcal{D}}, p_{\mathcal{D}}) \leftrightarrow (\mathcal{A}_q, \mathcal{A}_{\rm w})$ is algebraically invertible, so there is no remaining homogeneous perfect-fluid information after the atlas is known. The zero set of $\mathfrak{C}_I$ is exactly the constant-atlas condition, and the sign conditions are exactly $\epsilon_{\rm w,0}, \epsilon_q \geq 0$. Brown-current, four-form, topology, perturbation, and likelihood data do not alter this local algebraic classification; they determine whether the algebraic form is physically realized, amplitude-selected, observationally transferred, or statistically favored. Hence the listed ledgers are complete for the objects introduced by the standard, while making no claim about theories outside the standard's declared scope. □

**Definition 19** (Claim classes)**.** A manuscript-level statement is assigned to one of the following claim classes:

$$\begin{aligned} &\mathcal{C}_{\rm R}: \text{ residual reconstruction},\\ &\mathcal{C}_{\rm H}: \text{ homogeneous closure},\\ &\mathcal{C}_{\rm S}: \text{ source realization},\\ &\mathcal{C}_{\rm A}: \text{ amplitude selection},\\ &\mathcal{C}_{\rm P}: \text{ perturbative transfer},\\ &\mathcal{C}_{\rm G}: \text{ global causal/spectral topology},\\ &\mathcal{C}_{\rm L}: \text{ likelihood-grade evidence}. \end{aligned} \tag{32}$$

The standard forbids promoting a statement from $\mathcal{C}_i$ to a stronger class unless the additional invariant objects of that stronger class have been specified.

**Criterion 20** (Gold-standard success criterion)**.** *A dark-sector framework reaches gold-standard status only when it supplies a reproducible tuple*

$$\begin{aligned} \mathcal{G}_{\rm std} = (&\text{theorem},\ \text{audit table},\ \text{code interface},\\ &\text{public examples},\ \text{benchmark likelihood}). \end{aligned} \tag{33}$$

*The theorem prevents conceptual overclaiming, the table makes the claim class visible, the code interface makes closure defects reproducible, public examples show how standard models pass or fail, and the benchmark likelihood prevents confusing topology, amplitudes, and data-model choices.*

## IV. PLACEMENT IN THE CURRENT TOPOLOGY PROGRAM

The present result is deliberately narrower than a topology search. Current cosmic-topology work has moved from matched-circle exclusions toward harmonic covariance, quotient eigenmodes, observer-position dependence, tensor-mode covariance, and robustness against non-standard primordial spectra. Planck found no evidence for the specific compact topologies and anisotropic geometries tested in its matched-circle and Bayesian pipelines, principally flat toroidal or slab topologies and Bianchi VII$_h$ geometries; this is not an exclusion of all compact spherical quotients [15]. Spherical lens-space circle constraints must be read with their own $p, q$, curvature-scale, and observer-domain assumptions [17].

The COMPACT covariance program is used here as a methodological analogue unless a cited paper is explicitly spherical. Its Euclidean analyses show the required likelihood-level structure: matched-circle limits do not exhaust topology constraints, quotient boundary conditions alter eigenmodes and off-diagonal harmonic covariance, observer position can matter, tensor sectors have their own covariance, and non-standard primordial spectra can amplify or suppress detectability [18–22]. These statements support the ledger structure below; they are not used as a completed likelihood pipeline for every $S^3/\Gamma$.

Quantum-field effects on compact spaces define a separate stress-energy source. A topology-induced Casimir stress is not an amplitude selector for the minimal homogeneous branch. If such a stress produces anisotropic backreaction, it leaves the homogeneous locally-FLRW residual class rather than closing it from inside [23].

This paper therefore fixes the boundary of a background theorem. Its invariant input is only

$$(g_{\mu\nu},\ T^{\rm vis}_{\mu\nu},\ \text{Einstein-side convention},\ \Gamma),$$

and its output is a local stress-tensor normal form together with the conditions under which a compact quotient can

enter only through volume, spectra, covariance, and observer priors. It does not replace the covariance program; it supplies a convention-safe background ledger for it.

## V. EINSTEIN RESIDUAL AND CONVENTION COVARIANCE

Let $(M, g_{\mu\nu})$ be a four-dimensional spacetime, and let $T^{\rm vis}_{\mu\nu}$ denote the stress tensor assigned to the declared visible sector. Define

$$\boxed{T^{\mathcal{D}}_{\mu\nu} := \frac{c^4}{8\pi G} G_{\mu\nu} - T^{\rm vis}_{\mu\nu}.} \tag{34}$$

A bare cosmological-constant term is included in $T^{\mathcal{D}}_{\mu\nu}$ unless the Einstein-side convention is explicitly changed to put $G_{\mu\nu} + \Lambda g_{\mu\nu}$ on the left-hand side.

**Proposition 21** (Residual uniqueness and convention covariance)**.** *If*

$$G_{\mu\nu} = \frac{8\pi G}{c^4}\left(T^{\rm vis}_{\mu\nu} + S_{\mu\nu}\right), \tag{35}$$

*then $S_{\mu\nu} = T^{\mathcal{D}}_{\mu\nu}$. If the visible subtraction is changed by $X_{\mu\nu}$, then*

$$T^{\rm vis}_{\mu\nu} \to T^{\rm vis}_{\mu\nu} + X_{\mu\nu}, \qquad T^{\mathcal{D}}_{\mu\nu} \to T^{\mathcal{D}}_{\mu\nu} - X_{\mu\nu}. \tag{36}$$

*If the Einstein side is changed from $G_{\mu\nu}$ to $G_{\mu\nu} + \Lambda_0 g_{\mu\nu}$ while the visible subtraction is kept fixed, then*

$$T^{\mathcal{D}}_{\mu\nu} \to T^{\mathcal{D}}_{\mu\nu} + \frac{c^4\Lambda_0}{8\pi G} g_{\mu\nu}. \tag{37}$$

*This is a change in the definition of the Einstein-side residual, not the usual operation of moving a positive cosmological constant from the left-hand side to the right-hand side, where it is represented by $T^{\Lambda}_{\mu\nu} = -(c^4\Lambda/8\pi G) g_{\mu\nu}$.*

**Proposition 22** (Visible-subtraction covariance of the atlas)**.** *Suppose the homogeneous visible subtraction is changed by a homogeneous perfect-fluid tensor*

$$X_{\mu\nu} = \epsilon_X \frac{U_\mu U_\nu}{c^2} + p_X h_{\mu\nu}, \tag{38}$$

*so that $T^{\rm vis}_{\mu\nu} \to T^{\rm vis}_{\mu\nu} + X_{\mu\nu}$ and $T^{\mathcal{D}}_{\mu\nu} \to T^{\mathcal{D}}_{\mu\nu} - X_{\mu\nu}$. Then*

$$\epsilon_{\mathcal{D}} \to \epsilon_{\mathcal{D}} - \epsilon_X, \qquad p_{\mathcal{D}} \to p_{\mathcal{D}} - p_X, \tag{39}$$

*and the closure-pair coordinates transform affinely as*

$$\boxed{\mathcal{A}_q \to \mathcal{A}_q + p_X, \qquad \mathcal{A}_{\rm w} \to \mathcal{A}_{\rm w} - a^3(\epsilon_X + p_X).} \tag{40}$$

*Consequently, the statement that a residual is dust plus fixed vacuum is not an absolute statement about the metric alone. It is an invariant statement only after the Einstein-side convention and the visible-sector subtraction have been declared.*

We use standard differential-geometric conventions [1]. The contracted Bianchi identity gives

$$\nabla^\mu T^{\mathcal{D}}_{\mu\nu} = -\nabla^\mu T^{\rm vis}_{\mu\nu}. \tag{41}$$

Thus separate residual conservation is not automatic; it follows only when the visible sector is separately conserved in the adopted convention. This point matters because the minimal dust-plus-vacuum source branch is separately conserved.

We use a timelike four-velocity $U^\mu$ normalized by $U^\mu U_\mu = -c^2$ and the projector

$$h_{\mu\nu} = g_{\mu\nu} + \frac{U_\mu U_\nu}{c^2}. \tag{42}$$

For a perfect fluid with physical energy density $\epsilon$ and pressure $p$,

$$T_{\mu\nu} = \epsilon \frac{U_\mu U_\nu}{c^2} + p h_{\mu\nu}. \tag{43}$$

## VI. LOCALLY-FLRW RECONSTRUCTION

On the round cover, use

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + a^2(t) A_0^2 \mathrm{d}\Omega_3^2, \qquad \Omega_K = -\frac{c^2}{H_0^2 A_0^2} < 0, \tag{44}$$

where $A_0$ is the present curvature radius of the simply connected cover. For a quotient $S^3_{A_0}/\Gamma$, the local metric is unchanged while the volume and global spectrum are restricted. Let

$$E(a) = \frac{H(a)}{H_0}, \qquad \epsilon_{\rm crit,0} = \frac{3H_0^2 c^2}{8\pi G}. \tag{45}$$

The local Friedmann equations reconstruct a homogeneous residual as

$$\epsilon_{\mathcal{D}}(a) = \epsilon_{\rm crit,0}\left[E^2(a) - \Omega_K a^{-2}\right] - \epsilon_{\rm vis}(a), \tag{46}$$

$$p_{\mathcal{D}}(a) = -\frac{\epsilon_{\rm crit,0}}{3}\left[2E(a)\frac{\mathrm{d}E}{\mathrm{d}\ln a} + 3E^2(a) - \Omega_K a^{-2}\right] - p_{\rm vis}(a). \tag{47}$$

These are kinematical reconstructions from the chosen FLRW metric and subtraction convention. They do not assert residual field equations or any microscopic source model.

**Theorem 23** (Locally-FLRW residual reduction)**.** *After a fixed visible subtraction, every homogeneous and isotropic symmetric residual tensor is uniquely of the form*

$$T^{\mathcal{D}}_{\mu\nu} = \epsilon_{\mathcal{D}}(a) \frac{U_\mu U_\nu}{c^2} + p_{\mathcal{D}}(a) h_{\mu\nu}. \tag{48}$$

*If the residual is separately conserved, then*

$$\frac{\mathrm{d}\epsilon_{\mathcal{D}}}{\mathrm{d}\ln a} + 3(\epsilon_{\mathcal{D}} + p_{\mathcal{D}}) = 0. \tag{49}$$

*Proof.* Local homogeneity and isotropy leave only $U_\mu U_\nu$ and the spatial projector $h_{\mu\nu}$ as invariant symmetric rank-two tensors. Taking the covariant divergence in FLRW gives Eq. (49). □

## VII. CLOSURE-PAIR ATLAS AND MINIMAL BRANCH

Define the closure variables

$$\boxed{\mathcal{A}_q(a) := -p_{\mathcal{D}}(a), \qquad \mathcal{A}_{\rm w}(a) := a^3[\epsilon_{\mathcal{D}}(a) + p_{\mathcal{D}}(a)].} \tag{50}$$

They are a lossless reparametrization of the homogeneous residual:

$$p_{\mathcal{D}}(a) = -\mathcal{A}_q(a), \qquad \epsilon_{\mathcal{D}}(a) = a^{-3}\mathcal{A}_{\rm w}(a) + \mathcal{A}_q(a). \tag{51}$$

**Proposition 24** (Completeness of the homogeneous residual atlas)**.** *Within the homogeneous locally-FLRW perfect-fluid residual class, the pair $(\mathcal{A}_q, \mathcal{A}_{\rm w})$ is a complete set of coordinates. Any scalar diagnostic depending only on the homogeneous residual stress tensor and the declared scale factor factors through $(\mathcal{A}_q, \mathcal{A}_{\rm w})$. Therefore no additional homogeneous scalar criterion can distinguish a residual from dust plus fixed vacuum once $\mathcal{A}_q$ and $\mathcal{A}_{\rm w}$ are known; all further distinctions must be microscopic, perturbative, topological, or observational.*

If the residual is separately conserved, then

$$\frac{\mathrm{d}\mathcal{A}_q}{\mathrm{d}\ln a} + a^{-3}\frac{\mathrm{d}\mathcal{A}_{\rm w}}{\mathrm{d}\ln a} = 0. \tag{52}$$

Thus the general separately conserved homogeneous residual is a one-function flow in $(\mathcal{A}_q, \mathcal{A}_{\rm w})$ space. The minimal dust-plus-vacuum branch is the constant nonnegative submanifold.

**Theorem 25** (Homogeneous dust-plus-vacuum residual classification)**.** *Let $I \subset \mathbb{R}_+$ be a connected scale-factor interval, and let $(\epsilon_{\mathcal{D}}, p_{\mathcal{D}}) \in C^1(I)$ be the homogeneous locally-FLRW Einstein residual after a fixed visible-sector subtraction. Then, on $I$, the homogeneous local stress tensor is representable as*

$$T^{\mathcal{D}}_{\mu\nu} = \epsilon_{{\rm w},0}a^{-3}\frac{U_\mu U_\nu}{c^2} - \epsilon_q g_{\mu\nu}, \qquad \epsilon_{{\rm w},0}, \epsilon_q \geq 0, \tag{53}$$

*if and only if $\mathcal{A}_q$ and $\mathcal{A}_{\rm w}$ are constant and nonnegative on $I$. In that case*

$$\begin{aligned} \epsilon_q &= \mathcal{A}_q, \qquad & \epsilon_{{\rm w},0} &= \mathcal{A}_{\rm w}, \\ \epsilon_{\mathcal{D}}(a) &= \epsilon_{{\rm w},0}a^{-3} + \epsilon_q, \qquad & p_{\mathcal{D}}(a) &= -\epsilon_q. \end{aligned} \tag{54}$$

*Proof.* If Eq. (53) holds, then $-p_{\mathcal{D}} = \epsilon_q$ and $a^3(\epsilon_{\mathcal{D}} + p_{\mathcal{D}}) = \epsilon_{{\rm w},0}$. Conversely, if $\mathcal{A}_q$ and $\mathcal{A}_{\rm w}$ are constant, then Eq. (51) gives $p_{\mathcal{D}} = -\mathcal{A}_q$ and $\epsilon_{\mathcal{D}} = \mathcal{A}_{\rm w}a^{-3} + \mathcal{A}_q$, which is precisely Eq. (53). Nonnegativity is exactly the condition $\epsilon_q, \epsilon_{{\rm w},0} \geq 0$. □

Although this equivalence is algebraically elementary, its role is diagnostic: after the metric, Einstein-side convention, and visible subtraction are fixed, it gives the necessary and sufficient closure tests for interpreting the homogeneous residual as pressureless matter plus fixed vacuum. The theorem is therefore a convention-explicit closure diagnostic rather than a dynamical derivation. The "if and only if" is an iff statement for the homogeneous stress tensor. A microscopic branch realizes the constants only if its charge, mass scale, flux normalization, branch labels, and quotient volume admit those amplitudes. In particular, a discrete flux lattice can make some otherwise admissible constant values of $\epsilon_q$ unavailable unless the flux unit or branch spectrum is correspondingly specified.

The invariant null tests are

$$\mathcal{R}_q := \frac{\mathrm{d}\mathcal{A}_q}{\mathrm{d}\ln a} = 0, \qquad \mathcal{R}_{\rm w} := \frac{\mathrm{d}\mathcal{A}_{\rm w}}{\mathrm{d}\ln a} = 0, \qquad \mathcal{A}_q, \mathcal{A}_{\rm w} \geq 0. \tag{55}$$

For data or reconstruction pipelines it is useful to compress the same null condition into a nonnegative defect. For any fixed reference density $\epsilon_* > 0$, define

$$\mathfrak{C}_I[\mathcal{D}; \epsilon_*] := \sup_{a\in I} \frac{\left(\frac{\mathrm{d}\mathcal{A}_q}{\mathrm{d}\ln a}\right)^2 + \left(a^{-3}\frac{\mathrm{d}\mathcal{A}_{\rm w}}{\mathrm{d}\ln a}\right)^2}{\epsilon_*^2}. \tag{56}$$

The numerical value depends on the chosen norm and on $\epsilon_*$, but its zero set does not:

$$\begin{aligned} \mathfrak{C}_I[\mathcal{D}; \epsilon_*] = 0 &\iff \mathcal{R}_q = \mathcal{R}_{\rm w} = 0, \\ &\iff T^{\mathcal{D}}_{\mu\nu} \text{ has the minimal form on } I. \end{aligned} \tag{57}$$

Equivalently, a reconstruction pipeline is fully specified by the ordered map

$$\begin{aligned} (E(a), \epsilon_{\rm vis}, p_{\rm vis}, \Omega_K) &\longrightarrow (\epsilon_{\mathcal{D}}, p_{\mathcal{D}}) \\ &\longrightarrow (\mathcal{A}_q, \mathcal{A}_{\rm w}) \\ &\longrightarrow (\mathcal{R}_q, \mathcal{R}_{\rm w}, \mathfrak{C}_I). \end{aligned} \tag{58}$$

This makes the theorem operational: failures of closure are assigned to a specific visible subtraction and expansion history rather than to topology by default.

A failure of these tests falsifies the minimal homogeneous branch under the adopted visible subtraction; it does not falsify compact topology. No separate-conservation assumption is required for the algebraic iff statement. However, when the constants are realized, the resulting dust-plus-vacuum residual is separately conserved. Hence, by Eq. (41), a successful minimal decomposition also implies separate conservation of the declared visible sector in the same convention. Failure of visible-sector separate conservation is therefore already a failure of the adopted minimal source decomposition, not an independent topology effect.

**Corollary 26** (Local ΛCDM degeneracy)**.** *On any local FLRW chart, the minimal branch is exactly the stress*

*tensor of a pressureless component plus a cosmological-constant component with the same amplitudes and curvature. No homogeneous observable depending only on $H(a)$, distances, and the same curvature can distinguish the minimal branch from CDM plus $\Lambda$. A CDM plus $\Lambda$ model placed on the same compact quotient, with the same projected perturbation covariance, the same initial data, and the same amplitudes, remains degenerate with the minimal branch at this level.*

## VIII. CURRENT AND FOUR-FORM REPRESENTATION

One economical microscopic representation of the minimal stress tensor is a pressureless conserved-current branch plus a fixed four-form branch. The current branch may carry a protected degree label, but topology alone does not give a dust stress tensor.

The degree label is standard in algebraic topology [4], while the pressureless current representative is a Brown-fluid effective description [5]. A topological degree may label a sector, but by itself it does not imply a homogeneous dust current:

$$Q \in \mathbb{Z} \not\Rightarrow N^\mu = n(a)u^\mu \not\Rightarrow p_{\rm w} = 0. \tag{59}$$

Let $\Sigma$ be compact and oriented, let $\phi : \Sigma \to S^3$, and normalize the target volume form by $\int_{S^3} \widehat{\Omega} = 1$. The degree is

$$Q = \int_\Sigma \phi^* \widehat{\Omega} \in \mathbb{Z}. \tag{60}$$

The dust result requires a homogeneous coarse-grained Brown-current representative and a linear energy density

$$\epsilon_{\rm w}(n) = \mu_{\rm w} c^2 n. \tag{61}$$

More generally, a barotropic current effective theory with $\epsilon_J = F(n)$ has

$$p_J = nF'(n) - F(n), \qquad \frac{v_{s,J}^2}{c^2} := \frac{\mathrm{d}p_J}{\mathrm{d}\epsilon_J} = \frac{nF''(n)}{F'(n)}. \tag{62}$$

The minimal pressureless branch is the special case $F(n) = \mu_{\rm w} c^2 n$, after any additive vacuum offset has been removed and assigned to the vacuum sector. Then $p_J = 0$ and $v_{s,J} = 0$.

A compact-quotient specialization of the Brown current can be written as

$$S_{\rm J} = \frac{1}{c}\int_{\mathbb{R}\times\Sigma} \left[-\sqrt{-g}\, F(n) + \mathcal{J}^\mu\left(\partial_\mu \vartheta + \alpha_A \partial_\mu \beta^A\right)\right] \mathrm{d}^4x, \tag{63}$$

where $\mathcal{J}^\mu = \sqrt{-g}\, n u^\mu$ is the densitized current, $u^\mu u_\mu = -1$, and in this subsection $u^\mu := U^\mu/c$ is the dimensionless unit four-velocity compatible with $U^\mu U_\mu = -c^2$. Moreover,

$$n = \frac{1}{\sqrt{-g}}\sqrt{-g_{\mu\nu}\mathcal{J}^\mu \mathcal{J}^\nu}. \tag{64}$$

The Clebsch potentials enforce current conservation and advected labels; they do not add a topological amplitude. Variation gives the perfect-fluid form

$$\begin{aligned} T^{\rm J}_{\mu\nu} &= (\epsilon_{\rm J} + p_{\rm J}) u_\mu u_\nu + p_{\rm J} g_{\mu\nu}, \\ \epsilon_{\rm J} &= F(n), \qquad p_{\rm J} = nF'(n) - F(n). \end{aligned} \tag{65}$$

Thus the current action is specialized to a compact quotient once $(A_0, \Gamma, Q, F)$ are declared, but its amplitude still depends on the non-topological energy law $F$. The integer $Q$ labels a topological sector; it becomes a Brown-current number only after an additional branch rule is imposed.

**Criterion 27** (Current-EFT stability ledger)**.** *For a barotropic Brown-current realization of the residual, a conservative stability domain is*

$$\begin{aligned} F(n) \geq 0, \qquad F'(n) > 0, \\ 0 \leq \frac{v_s^2}{c^2} = \frac{\mathrm{d}p}{\mathrm{d}\epsilon} = \frac{nF''(n)}{F'(n)} \leq 1. \end{aligned} \tag{66}$$

*The first two conditions give positive energy and positive inertial enthalpy, $\epsilon + p = nF'(n) > 0$; the last condition gives nonnegative and subluminal adiabatic response. Pure pressureless current behavior requires the special stable line*

$$F(n) = \mu_{\rm w} c^2 n, \qquad p_{\rm w} = 0, \qquad v_{s,{\rm w}}^2 = 0, \tag{67}$$

*after any additive vacuum offset has been removed and assigned consistently to the vacuum sector. More generally, $F(n) = \mu_{\rm w} c^2 n + C$ gives $p_J = -C$, namely dust plus a vacuum counterterm rather than a pure current. Any nonlinear $F$ is a different current EFT and must pass the same closure and perturbative null tests rather than being identified with the minimal branch.*

Equivalently, the stable barotropic current class may be generated by a function

$$0 \leq s(n) \leq 1, \qquad s(n) := \frac{\mathrm{d}\ln F'(n)}{\mathrm{d}\ln n}, \tag{68}$$

which gives

$$F'(n) = F'(n_0)\exp\left[\int_{n_0}^{n} s(x)\,\mathrm{d}\ln x\right], \tag{69}$$

$$F(n) = F(n_0) + \int_{n_0}^{n} F'(y)\,\mathrm{d}y. \tag{70}$$

The case $s(n) = 0$ gives $F(n) = \mu c^2 n + \text{const}$, i.e. dust plus a vacuum offset; the offset must be assigned to the vacuum sector to avoid double counting with the four-form branch. The case $s(n) = w$ with $0 < w \leq 1$ gives $F(n) \propto n^{1+w}$ and $p = w\epsilon$; it is stable but not the minimal dust residual.

To use the degree as the Brown-current number one must impose the sector identification

$$N_J := \int_{\Sigma_a} n\,\mathrm{d}V_a = |Q|. \tag{71}$$

This is an additional branch assignment, not a consequence of the degree definition alone: $Q \in \mathbb{Z}$ labels a topological sector, and only the extra branch rule $N_J = |Q|$ turns it into a Brown-current number. With this assignment and homogeneous coarse graining on $\Sigma_{A_0} = S^3_{A_0}/\Gamma$,

$$\mathrm{Vol}(\Sigma_{A_0}) = \frac{2\pi^2 A_0^3}{|\Gamma|}, \qquad n(a) = \frac{|Q|}{a^3\,\mathrm{Vol}(\Sigma_{A_0})}, \tag{72}$$

and therefore

$$\Omega_{\mathrm{w}} = \frac{\mu_{\mathrm{w}} c^2 |Q|}{\mathrm{Vol}(\Sigma_{A_0})\epsilon_{\mathrm{crit},0}} = \frac{4G\mu_{\mathrm{w}}|Q|\,|\Gamma|}{3\pi H_0^2 A_0^3}. \tag{73}$$

This is an exact bookkeeping identity only after the branch postulate $N_J = |Q|$ and the non-topological data $(\mu_{\mathrm{w}}, |\Gamma|, A_0)$ are supplied. It is not an amplitude prediction.

Four-form vacuum sectors of this kind are standard in gravitational theory [6–9]. For a four-form $F_4 = \mathrm{d}C_3$, take

$$S_F = -\frac{Z_F}{2\cdot 4!}\int \sqrt{-g}\, F_{\mu\nu\rho\sigma}F^{\mu\nu\rho\sigma}\,\mathrm{d}^4x, \qquad Z_F > 0. \tag{74}$$

The stress tensor obtained from Eq. (74) is

$$T^F_{\mu\nu} = \frac{Z_F}{3!}F_{\mu\alpha\beta\gamma}F_\nu{}^{\alpha\beta\gamma} - \frac{Z_F}{2\cdot 4!}g_{\mu\nu}F_{\alpha\beta\gamma\delta}F^{\alpha\beta\gamma\delta}. \tag{75}$$

The field equation implies that the four-form is proportional to the metric volume form on each connected branch,

$$F_{\mu\nu\rho\sigma} = f\varepsilon_{\mu\nu\rho\sigma}, \qquad f = \text{constant}, \tag{76}$$

up to the orientation convention for the Hodge dual. Only $f^2$ enters the stress tensor. Define

$$\widehat{q} = \sqrt{Z_F}f, \qquad \epsilon_q = \frac{1}{2}\widehat{q}^2. \tag{77}$$

With the $(-+++)$ signature,

$$\varepsilon_{\mu\alpha\beta\gamma}\varepsilon_\nu{}^{\alpha\beta\gamma} = -3!g_{\mu\nu}, \qquad F_{\alpha\beta\gamma\delta}F^{\alpha\beta\gamma\delta} = -4!f^2, \tag{78}$$

which gives

$$T^q_{\mu\nu} = -\epsilon_q g_{\mu\nu}, \qquad \Lambda_{\mathrm{eff}} = \frac{8\pi G}{c^4}\epsilon_q. \tag{79}$$

**Proposition 28** (Three separate four-form statements)**.** For the four-form model considered here, three logically distinct statements must not be conflated. *First,*

$$F_4 = f\,\varepsilon_4 \quad \Rightarrow \quad T^q_{\mu\nu} = -\epsilon_q g_{\mu\nu}, \qquad \epsilon_q = \frac{1}{2}\widehat{q}^2. \tag{80}$$

*This is local stress-tensor closure on a connected fixed branch and follows from Eqs. (75)–(79). Second,*

$$\epsilon_q = \frac{1}{2}\widehat{q}^2, \qquad \widehat{q} \in \mathbb{R}, \tag{81}$$

*is a continuous amplitude statement. Third,*

$$\widehat{q} = N_F e_F, \qquad N_F \in \mathbb{Z}, \tag{82}$$

*is an additional flux-quantization rule. The third statement is not implied by compact spatial topology alone; it requires a charge normalization, compact zero-mode normalization, membrane sector, or equivalent branch rule.*

For a fixed superselection branch, the mixed-index perturbation vanishes,

$$\delta T^q{}_\mu{}^\mu{}_\nu = 0, \tag{83}$$

while the covariant-index variation

$$\delta T^q_{\mu\nu} = -\epsilon_q \delta g_{\mu\nu} \tag{84}$$

is purely metric-induced.

**Proposition 29** (Closure versus flux-lattice selection)**.** *The homogeneous closure criterion determines only the stress-energy form*

$$T^q_{\mu\nu} = -\epsilon_q g_{\mu\nu}, \qquad \epsilon_q = \textit{constant} \geq 0. \tag{85}$$

*It does not by itself determine whether the constant belongs to a discrete flux lattice. Spatial compactness by itself does not quantize the four-form amplitude; the lattice enters only after a charge normalization, compact zero-mode normalization, or membrane sector has been specified. If such a normalization is supplied so that*

$$\widehat{q} = N_F e_F, \qquad N_F \in \mathbb{Z}, \tag{86}$$

*then the admissible vacuum amplitudes are restricted to*

$$\epsilon_q = \frac{1}{2}N_F^2 e_F^2. \tag{87}$$

*Thus the FLRW closure theorem is a continuous stress-energy classification, whereas flux quantization is an additional amplitude-selection constraint. Arbitrary residual matching requires an unfixed normalization $e_F$, a dense landscape of allowed branches, or a phenomenological effective constant.*

For an observational target $\epsilon_q^{\mathrm{target}}$ and a single flux unit $e_F$, the corresponding branch selector is the lattice problem

$$N_F^\star \in \operatorname*{argmin}_{N_F\in\mathbb{Z}} \left|\frac{1}{2}N_F^2 e_F^2 - \epsilon_q^{\mathrm{target}}\right|. \tag{88}$$

Exact fixed-vacuum closure requires

$$\frac{1}{2}(N_F^\star)^2 e_F^2 = \epsilon_q^{\mathrm{target}}, \tag{89}$$

whereas phenomenological closure only requires the mismatch to lie below the adopted observational tolerance. For the fixed-four-form model considered here, the amplitude routes are a continuous EFT branch $q_b \in \mathbb{R}$, a single discrete flux $q_b = N_F e_F$, a dense multiflux lattice $\epsilon_q = \frac{1}{2}\sum_I N_I^2 e_I^2$, a directly fitted phenomenological constant, or no admissible fixed branch.

**Criterion 30** (Fixed four-form branch)**.** *On a scale-factor interval $I$, the fixed-vacuum source law requires*

$$\frac{\mathrm{d}N_F}{\mathrm{d}\ln a}=0, \qquad \frac{\mathrm{d}e_F}{\mathrm{d}\ln a}=0, \qquad \text{no membrane transition on } I. \tag{90}$$

*This can be imposed either as strict superselection,*

$$\mathcal{H}=\bigoplus_{N_F}\mathcal{H}_{N_F}, \qquad |\Psi_{\cos}\rangle\in\mathcal{H}_{N_F^\star}, \tag{91}$$

*or as EFT transition suppression,*

$$\Gamma_{\mathrm{mem}}\,\mathcal{V}_4(I)\ll 1. \tag{92}$$

*If membrane transitions are allowed without suppression in a flux-lattice branch, then*

$$N_F\longrightarrow N_F\pm n_{\mathrm{mem}}, \qquad n_{\mathrm{mem}}\in\mathbb{Z}, \tag{93}$$

*so that*

$$\epsilon_q:\frac{1}{2}N_F^2e_F^2\longrightarrow\frac{1}{2}(N_F\pm n_{\mathrm{mem}})^2e_F^2, \tag{94}$$

*and the theorem applies only piecewise between jumps; the residual is not globally fixed vacuum.*

## IX. MINIMAL BACKGROUND SYSTEM AND COMPACT CAUSAL LEDGER

The minimal background system represented by the closed homogeneous atlas is

$$G_{\mu\nu}=\frac{8\pi G}{c^4}\left(T^{\mathrm{vis}}_{\mu\nu}+T^{\mathrm{w}}_{\mu\nu}+T^{q}_{\mu\nu}\right), \tag{95}$$

with

$$T^{\mathrm{w}}_{\mu\nu}=\epsilon_{\mathrm{w}}\frac{U_\mu U_\nu}{c^2}, \qquad \epsilon_{\mathrm{w}}=\mu_{\mathrm{w}}c^2n, \qquad \nabla_\mu(nU^\mu/c)=0, \tag{96}$$

and

$$T^{q}_{\mu\nu}=-\epsilon_q g_{\mu\nu}. \tag{97}$$

For the closed round cover this gives the background Friedmann relation

$$H^2+\frac{c^2}{a^2A_0^2}=\frac{8\pi G}{3c^2}\left(\epsilon_{\mathrm{vis}}+\epsilon_{\mathrm{w},0}a^{-3}+\epsilon_q\right). \tag{98}$$

Equation (98) is not an amplitude derivation. It is the background equation after the constants $(\epsilon_{\mathrm{w},0},\epsilon_q,A_0)$ and the visible sector have been supplied.

The compact causal part of the framework is source-free. On the simply connected closed cover

$$\mathrm{d}s^2=-c^2\mathrm{d}t^2+a^2(t)A_0^2\mathrm{d}\Omega_3^2, \tag{99}$$

null propagation defines the dimensionless conformal causal distance

$$\Delta\chi(t_1,t_2)=\int_{t_1}^{t_2}\frac{c\,\mathrm{d}t}{a(t)A_0}. \tag{100}$$

Because the geodesic diameter of the unit three-sphere is $\pi$, full causal reach on the cover requires

$$\Delta\chi(t_1,t_2)\geq\pi. \tag{101}$$

For a static cover $a=1$, this reduces to

$$t_{\mathrm{coh}}=\frac{\pi A_0}{c}. \tag{102}$$

The corresponding cover-volume accessibility ratio is

$$\mathcal{G}_{\mathrm{cau}}(t_1,t_2)=\frac{\mathrm{Vol}\,B_\chi}{\mathrm{Vol}\,S^3}, \qquad \mathcal{G}_{\mathrm{cau}}(t_1,t_2)=\frac{\chi-\frac{1}{2}\sin(2\chi)}{\pi}, \qquad \chi=\min\{\Delta\chi,\pi\}. \tag{103}$$

This is the invariant content of the compact causal ledger on the simply connected cover: light supplies null cones, the compact cover supplies a finite causal diameter, and expansion enters only through conformal time. The causal ledger does not modify Eq. (95) and does not select $\epsilon_{\mathrm{w},0}$ or $\epsilon_q$.

For a quotient $\Sigma_\Gamma=S^3/\Gamma$, Eq. (100) remains the local null-reach variable on the cover, but the exact accessibility condition is quotient-relative. For an observer $x_o$, define

$$D_\Gamma(x_o):=\sup_{y\in\Sigma_\Gamma}d_\Gamma(x_o,y), \tag{104}$$

where $d_\Gamma$ is the quotient geodesic distance. Full causal access to the quotient from $x_o$ is equivalent to

$$\Delta\chi(t_1,t_2)\geq D_\Gamma(x_o), \tag{105}$$

and the invariant accessibility fraction is

$$\mathcal{G}^\Gamma_{\mathrm{cau}}(x_o;t_1,t_2)=\frac{\mathrm{Vol}\{y\in\Sigma_\Gamma:d_\Gamma(x_o,y)\leq\Delta\chi\}}{\mathrm{Vol}(\Sigma_\Gamma)}. \tag{106}$$

For the simply connected cover, $D_\Gamma=\pi$ and Eq. (106) reduces to Eq. (103). Thus compact causal closure requires the same data as compact spectral closure: the quotient $\Gamma$, its projector, and the observer-position prior. A model with no specified topology is locally viable but globally incomplete.

**Criterion 31** (Causal ledger is not a source)**.** *Compact causal accessibility is a propagation and global-identification criterion. It may constrain which regions or repeated images are mutually causally accessible, and it may enter topology-covariance calculations through the quotient geometry. It is not a local homogeneous stress tensor and cannot replace the Brown-current mass scale, the four-form flux unit, or the compact Boltzmann transfer map.*

**Proposition 32** (Local Euclidean limit)**.** *Let $L$ be the physical size of a laboratory or survey patch with $L \ll aA_0$. In normal coordinates on the spatial slice, the compact spherical metric differs from the Euclidean FLRW metric by relative corrections of order $L^2/(a^2A_0^2)$. Therefore the residual atlas, Brown-current dust law, and fixed four-form vacuum law are locally the same EFT statements as in an $R^3$ FLRW chart. Compactness first appears through global identifications, spectra, causal reach, and covariance, not through a different local stress tensor.*

## X. COMPACT TOPOLOGY AND QUOTIENT PROJECTION

The spatial topology is a global input, not a local equation of state. The local residual classification above did not require compactness or positive curvature. The global ledger in this section does: throughout the topological-current and compact three-form slots we restrict to compact orientable spherical space forms, equivalently to finite freely acting orientation-preserving isometry groups $\Gamma$ on $S^3$. Without orientability the degree and top-cohomology statements must be replaced by their twisted-coefficient versions; such branches are not considered here. The classification of complete constant-curvature space forms gives [2, 3]

$$\Sigma = \widetilde{\Sigma}/\Gamma, \qquad \widetilde{\Sigma} \in \{S^3, \mathbb{R}^3, H^3\}. \tag{107}$$

In the compact positive-curvature case,

$$\Sigma = S^3_A/\Gamma, \qquad \mathrm{Vol}(\Sigma) = \frac{2\pi^2 A^3}{|\Gamma|}. \tag{108}$$

If simple connectivity is assumed, $\Gamma = \mathbf{1}$ and $\Sigma \simeq S^3_A$. Without simple connectivity, the invariant conclusion is only that the universal cover is $S^3_A$.

On the simply connected cover,

$$-\Delta_{S^3_A} Y_{n\ell m} = \frac{n(n+2)}{A^2} Y_{n\ell m}, \qquad \mathrm{mult}(n) = (n+1)^2. \tag{109}$$

For a quotient, fields are $\Gamma$-invariant cover fields. The projector is

$$\boxed{P_\Gamma = \frac{1}{|\Gamma|} \sum_{\gamma \in \Gamma} \gamma^*.} \tag{110}$$

The physical scalar eigenspace is $\mathcal{H}_n^\Gamma = P_\Gamma \mathcal{H}_n$, with multiplicity

$$m_\Gamma(n) = \mathrm{Tr}_{\mathcal{H}_n} P_\Gamma. \tag{111}$$

The same logic applies to physical vector and tensor harmonic sectors after closed-FLRW gauge and constraint modes are removed.

Compact topology can therefore enter through

$$\mathrm{Vol}(\Sigma), \qquad \mathrm{Spec}_\Gamma(\Delta), \qquad m_\Gamma(n), \qquad \mathrm{Cov}_\Gamma, \tag{112}$$

not through a new local homogeneous stress tensor. Observer-position dependence appears for non-globally-homogeneous quotients; for globally homogeneous quotients the covariance is still quotient-projected but is not position dependent in the same way.

**Criterion 33** (Topology-evidence boundary)**.** *A compact spherical model may claim compact-background consistency after specifying $S^3/\Gamma$ and the quotient projection. It may not claim topology evidence or benchmark superiority until the projector has been propagated into transfer functions and observable covariance,*

$$P_\Gamma \longrightarrow m_\Gamma(n),\ \Xi_n^\Gamma,\ \Delta_{\ell n}^{X,\Gamma},\ C_\ell^{XY,\Gamma},\ \mathrm{Cov}_\Gamma,\ \mathcal{L}_\Gamma. \tag{113}$$

*A background quotient without this propagated covariance is a compact background ansatz, not a topology likelihood.*

## XI. AMPLITUDE-SELECTION BOUNDARY

Compact topology supplies labels, volume, and spectral restrictions. It does not supply numerical cosmological abundances. In particular,

$$(\Gamma, [\Sigma, S^3], H^3(\Sigma, \mathbb{Z}), Q, N_F) \not\Rightarrow (\Omega_{\mathrm{w}}, \Omega_q, \Lambda_{\mathrm{eff}}). \tag{114}$$

Even after the geometric radius $A_0$ is supplied, the amplitudes require $\mu_{\mathrm{w}}$, $e_F$, branch labels, compact-gauge normalization, and a selection rule or phenomenological prior. This is an amplitude-selection boundary, not a failure of the closure theorem.

**Theorem 34** (Topological decoupling of amplitudes)**.** *Fix the class of compact orientable spherical space forms and the discrete sector data*

$$\mathcal{T}_{\mathrm{top}} = (\Gamma, [\Sigma, S^3], H^3(\Sigma, \mathbb{Z}), Q, N_F).$$

*There is no convention-independent, dimensionally complete topology-only map from $\mathcal{T}_{\mathrm{top}}$ to the physical amplitude pair $(\Omega_{\mathrm{w}}, \Omega_q)$. Any physical amplitude selector must factor through additional non-topological data,*

$$\mathfrak{S}_{\mathrm{amp}} : (\mathcal{T}_{\mathrm{top}}; A_0, H_0, \mu_{\mathrm{w}}, e_F, \mathcal{B}) \longrightarrow (\Omega_{\mathrm{w}}, \Omega_q), \tag{115}$$

*where $\mathcal{B}$ denotes branch, compact-gauge, boundary, or prior data. Equivalently, the microscopic selector splits into independent current and vacuum slots,*

$$\mathcal{K}_{\mathrm{micro}} = \mathcal{K}_{\mathrm{w}} \oplus \mathcal{K}_q, \qquad \mathcal{K}_{\mathrm{w}} \ni \mu_{\mathrm{w}}, \qquad \mathcal{K}_q \ni e_F. \tag{116}$$

*Thus topology fixes admissible sectors and projected spectra, whereas microphysics or phenomenological branch selection fixes amplitudes.*

*Proof.* Holding $(\Gamma, Q, N_F)$ fixed while varying $\mu_{\mathrm{w}}$, $e_F$, $A_0$, or $H_0$ changes $(\Omega_{\mathrm{w}}, \Omega_q)$. Hence the amplitudes cannot be functions of topology alone. Equivalently, the entries in

$\mathcal{T}_{\rm top}$ label sectors and restrict mode spaces, but they contain no conversion from charge number to energy density. For the current branch,

$$\Omega_{\rm w} = \frac{\mu_{\rm w} c^2 |Q|}{{\rm Vol}(\Sigma_{A_0})\epsilon_{{\rm crit},0}}, \qquad {\rm Vol}(\Sigma_{A_0}) = \frac{2\pi^2 A_0^3}{|\Gamma|}. \quad (117)$$

Therefore $Q$, $\Gamma$, and the topological class do not determine $\Omega_{\rm w}$ unless the effective mass scale $\mu_{\rm w}$, the geometric scale $A_0$, and the critical-density normalization are supplied. For the four-form branch,

$$\Omega_q = \frac{\epsilon_q}{\epsilon_{{\rm crit},0}}, \qquad \epsilon_q = \frac{1}{2} N_F^2 e_F^2. \quad (118)$$

Therefore the flux integer $N_F$ does not determine $\Omega_q$ unless the flux unit $e_F$ and the same cosmological normalization are supplied. Consequently, any numerical map from topological labels alone to $(\Omega_{\rm w}, \Omega_q)$ is either a phenomenological assignment or a hidden use of additional microphysical or boundary data. This proves the decoupling statement. □

A useful diagnostic for selector strength is the curvature-linear ansatz $\Omega_{\rm w} = C|\Omega_K|$. If observations impose $|\Omega_K| < \kappa$ and the target is $\Omega_\star$, then

$$C > \frac{\Omega_\star}{\kappa}. \quad (119)$$

For $\Omega_\star \simeq 0.26$ and $\kappa = 0.005$, one needs $C > 52$. A toy holographic winding saturation condition,

$$\mu_{\rm w} c^2 |Q| = \frac{c^4 A_0}{2G}, \quad (120)$$

gives only

$$\Omega_{\rm w}^{\rm HWS} = \frac{2}{3\pi}|\Omega_K| \quad (121)$$

for the simply connected case. For a quotient, imposing the same total-energy saturation on the quotient volume gives

$$\Omega_{\rm w}^{\rm HWS}(\Gamma) = \frac{2|\Gamma|}{3\pi}|\Omega_K|. \quad (122)$$

This is not a derivation of $\Omega_c$. If $|\Gamma| \sim 10^2$ is used to obtain a large effective coefficient, the proposal becomes a large-quotient amplitude ansatz that must pass matched-circle, observer-position, covariance, and spectral-multiplicity checks.

## XII. PERTURBATIVE AND OBSERVATIONAL NULL TESTS

We use standard gauge-invariant cosmological perturbation conventions [10–13]. At linear order, the residual is defined before gauge choice by

$$\delta T^{\mathcal{D}}_{\mu\nu} = \frac{c^4}{8\pi G}\delta G_{\mu\nu} - \delta T^{\rm vis}_{\mu\nu}. \quad (123)$$

The fixed four-form branch contributes no independent scalar perturbation on a fixed branch. After subtracting this fixed vacuum component, the minimal current branch must obey the dust null conditions on every physical quotient-projected scalar mode:

$$\boxed{\delta p_{\rm w,rf} = 0, \qquad \Pi_{\rm w}^{\rm GI} = 0, \qquad \frac{v_{s,\rm w}^2}{c^2} = 0.} \quad (124)$$

A statistically significant intrinsic pressure perturbation, rest-frame sound speed, anisotropic stress, entropy source, or independent non-dust stress source not assigned to the visible sector falsifies the minimal perturbative branch.

Current background data sharpen the closure tests without changing their logical status. Planck 2018 combined with BAO gives a curvature constraint consistent with flatness, conventionally quoted as $\Omega_K = 0.0007 \pm 0.0019$ for TT,TE,EE+lowE+lensing+BAO [16]. Recent DESI DR2 BAO results are well described by flat ΛCDM, while BAO+CMB prefers a $w_0 w_a$ model over ΛCDM at $3.1\sigma$, and BAO+CMB plus supernova samples gives a sample-dependent preference in the range $2.8$–$4.2\sigma$ [24]. If a reconstructed residual pressure evolution remains significant after systematics, dataset tension, and visible-sector convention dependence are accounted for, then the fixed-four-form minimal branch fails the $\mathcal{R}_q = 0$ null test. That would falsify the fixed-vacuum source law, not compact spherical topology.

Likewise, matched-circle and compact-covariance limits constrain detectable domains and topology families, not every compact spherical quotient. The published lens-space constraints are strong statements for specified $L(p,q)$, curvature-scale ranges, and circle-search domains; they are not a completed scan of all $S^3/\Gamma$ likelihoods. Lens-space constraints are therefore constraints on specified $S^3/\Gamma$ implementations and their observer-position/covariance pipelines, where applicable [14, 15, 17].

This separation also appears in compact-topology covariance studies. The quotient changes the allowed eigenmodes and covariance. Assumptions about large-scale primordial power can amplify or suppress detectability of the quotient signal [19–22]. A Casimir stress is a different physical source, not an amplitude selector for the minimal homogeneous branch. If it produces anisotropic backreaction, it exits the homogeneous locally-FLRW residual class rather than selecting the amplitudes inside it [23].

## XIII. WHAT WOULD BE REQUIRED FOR A TOPOLOGY-LIKELIHOOD CLAIM

The following section is not part of the homogeneous closure theorem. It records the additional ingredients required before a compact-background ansatz could be promoted to a topology-evidence calculation. The homogeneous theorem is a background classification: a topology-evidence claim begins only after the quotient

projector is propagated to the covariance of an actual data vector. The admissible completion is therefore a protocol, not a detection claim. It requires a concrete quotient $\Gamma$, a data vector, a mask, a foreground and noise model, a compact Boltzmann implementation, and a likelihood comparison.

### A. SVT harmonic ledger

Let $r = s, v, t$ denote scalar, transverse-vector, and transverse-traceless tensor sectors after gauge and constraint modes are removed. Quotient harmonics are obtained by projection from the cover,

$$P_\Gamma = \frac{1}{|\Gamma|}\sum_{\gamma\in\Gamma}\gamma^*, \qquad Y^\Gamma_{n\alpha} = P_\Gamma Y_{n\alpha}, \tag{125}$$

and analogously for $V^\Gamma_{n\alpha i}$ and $T^\Gamma_{n\alpha ij}$. Because scalar, vector, and tensor spectra need not share the same eigenvalue formula, write

$$-\Delta_r Q^{(r),\Gamma}_{n\alpha} = \lambda^{(r)}_n Q^{(r),\Gamma}_{n\alpha}, \qquad r = s, v, t. \tag{126}$$

The quotient multiplicities are

$$m^{(r)}_\Gamma(n) = \dim\ker\big(\Delta_r + \lambda^{(r)}_n\big)^\Gamma, \qquad r = s, v, t. \tag{127}$$

Here the superscript $\Gamma$ means the invariant subspace selected by $P_\Gamma$.

In Newtonian gauge,

$$\mathrm{d}s^2 = a^2(\eta)\{-(1+2\Psi)\mathrm{d}\eta^2 + (1-2\Phi)\gamma_{ij}\mathrm{d}x^i\mathrm{d}x^j\}. \tag{128}$$

For the minimal dust current one may write, up to the sign convention for scalar harmonics and metric potentials,

$$\delta'_{\mathrm{w},n\alpha} = -\theta_{\mathrm{w},n\alpha} + 3\Phi'_{n\alpha}, \tag{129}$$

$$\theta'_{\mathrm{w},n\alpha} + \mathcal{H}\theta_{\mathrm{w},n\alpha} = \lambda^{(s)}_n \Psi_{n\alpha}. \tag{130}$$

The necessary null requirements are

$$\delta p_{\mathrm{w,rf}} = 0, \qquad \Pi^{\mathrm{GI}}_{\mathrm{w}} = 0, \qquad v^2_{s,\mathrm{w}} = 0. \tag{131}$$

For a stable non-dust current EFT, a schematic scalar ledger is

$$\delta'_{J,n\alpha} = -(1+w_J)(\theta_{J,n\alpha} - 3\Phi'_{n\alpha}) - 3\mathcal{H}(c_s^2 - w_J)\delta_{J,n\alpha}, \tag{132}$$

$$\theta'_{J,n\alpha} = -\mathcal{H}(1-3c_s^2)\theta_{J,n\alpha} - \frac{w'_J}{1+w_J}\theta_{J,n\alpha} \tag{133}$$

$$+ \frac{c_s^2}{1+w_J}\lambda^{(s)}_n\delta_{J,n\alpha} + \lambda^{(s)}_n\Psi_{n\alpha}. \tag{134}$$

Vector and tensor null conditions for the residual are

$$\Pi^{(v)}_{\mathrm{w},n\alpha} = 0, \qquad v^{(v)\prime}_{\mathrm{w},n\alpha} + \mathcal{H}v^{(v)}_{\mathrm{w},n\alpha} = 0, \tag{135}$$

$$\Pi^{(t)}_{\mathrm{w},n\alpha} = 0, \qquad \Pi^{(t)}_{q,n\alpha} = 0. \tag{136}$$

For tensor perturbations, use the kinetic operator actually appearing in the linearized Einstein equation on the compact quotient,

$$\mathcal{K}_t Q^{(t),\Gamma}_{n\alpha} = \kappa^{(t)}_n Q^{(t),\Gamma}_{n\alpha}. \tag{137}$$

This convention-safe notation absorbs the curvature shift that would otherwise depend on whether one calls the eigenvalue a rough-Laplacian, Lichnerowicz, or already-shifted kinetic eigenvalue. The tensor amplitudes obey schematically

$$h''_{n\alpha} + 2\mathcal{H}h'_{n\alpha} + \kappa^{(t)}_n h_{n\alpha} = \frac{16\pi G}{c^4}a^2\Pi^{(t)}_{\mathrm{vis},n\alpha}. \tag{138}$$

For the four-form sector, on a fixed branch,

$$\delta T^{q\,\mu}{}_\nu = 0, \qquad \delta T^q_{\mu\nu} = -\epsilon_q \delta g_{\mu\nu}. \tag{139}$$

Thus the branch is unperturbed as a physical vacuum component; the covariant index variation is metric-induced.

### B. Compact transfer functions and covariance

For an observable $X \in \{T, E, B, \phi_{\mathrm{lens}}, g, \ldots\}$ define

$$a^X_{\ell m}(x_o, R) = \sum_{n,\alpha,I}\Delta^{X,\Gamma,I}_{\ell m;n\alpha}(x_o, R; \eta)\mathcal{R}^I_{n\alpha}. \tag{140}$$

Here $x_o$ is the observer position in $\Sigma_\Gamma$, $R \in SO(3)$ is sky orientation, and $I$ labels scalar, vector, tensor, or other declared initial sectors. The quotient-projected primordial covariance is

$$\langle \mathcal{R}^I_{n\alpha}\mathcal{R}^{J*}_{n'\beta}\rangle = \Xi^{IJ,\Gamma}_{n\alpha,n'\beta}. \tag{141}$$

Hence, using multi-indices $A = (n, \alpha, I)$ and $B = (n', \beta, J)$,

$$\mathrm{Cov}^{XY,\Gamma}_{\ell m,\ell' m'}(x_o, R) = \sum_{A,B}\Delta^{X,\Gamma}_{\ell m;A}\,\Xi^\Gamma_{AB}\,\Delta^{Y,\Gamma*}_{\ell' m';B}. \tag{142}$$

The required chain is

$$P_\Gamma \longrightarrow m^{(r)}_\Gamma(n) \longrightarrow \Xi^\Gamma \longrightarrow \Delta^{X,\Gamma} \longrightarrow \mathrm{Cov}^\Gamma \longrightarrow \mathcal{L}_\Gamma. \tag{143}$$

At the perturbative level, the exact degeneracy statement is

$$\begin{aligned}&\text{same } (\epsilon_{\mathrm{w},0}, \epsilon_q), \quad \text{same visible sector,}\\ &\text{same } \Xi^\Gamma, \quad \text{same } \Delta^{X,\Gamma} \Longrightarrow \text{same } \mathrm{Cov}^{XY,\Gamma},\\ &\Longrightarrow \text{no observable distinction from CDM} + \Lambda.\end{aligned} \tag{144}$$

within the declared observable data model. Initial statistics and transfer functions determine the covariance; the covariance determines the likelihood. A background quotient without the propagated covariance in Eq. (143) is a compact background ansatz, not a topology likelihood.

### C. Observer-position priors and likelihood

For globally homogeneous quotients the observer-position integral collapses to one orbit. For non-globally-homogeneous quotients one must integrate over the inequivalent observer domain

$$\mathcal{F}_\Gamma = \Sigma_\Gamma/\mathrm{Isom}(\Sigma_\Gamma). \tag{145}$$

The default invariant prior is volume-uniform,

$$\pi_\Gamma(x_o) = \frac{1}{\mathrm{Vol}(\mathcal{F}_\Gamma)}, \qquad \int_{\mathcal{F}_\Gamma} \pi_\Gamma \, \mathrm{d}\mu_\Gamma = 1. \tag{146}$$

More general priors may be used only if declared before fitting,

$$\pi_\Gamma(x_o|\mathcal{S}) \propto \pi_{\mathrm{geom}}(x_o) W_{\mathrm{selection}}(x_o). \tag{147}$$

Hidden observer tuning is not an admissible topology claim.

For a fixed $(\Gamma, x_o, R, \eta)$ the masked data covariance is written as

$$C_\Gamma = M \, \mathrm{Cov}_\Gamma(x_o, R, \eta) \, M^\dagger + N + C_{\mathrm{fg}}(\nu). \tag{148}$$

Here $M$ is the mask or mode-coupling operator, $N$ is instrumental noise, and $\nu$ denotes declared foreground nuisance parameters. The covariance-level Gaussian likelihood is

$$\mathcal{L}_\Gamma(d) = \int_{\mathcal{F}_\Gamma} \pi_\Gamma(x_o) \, \mathrm{d}\mu_\Gamma(x_o) \int_{SO(3)} \mathrm{d}R \times \int \mathrm{d}\eta \, \mathrm{d}\nu \; \mathcal{G}(d; C_\Gamma) \Pi(\eta, \nu), \tag{149}$$

$$\mathcal{G}(d; C) = \frac{\exp[-\frac{1}{2} d^\dagger C^{-1} d]}{\sqrt{\det(2\pi C)}}. \tag{150}$$

The comparison model must use the same local expansion history, curvature, visible sector, background amplitudes, mask, noise, and foreground conventions, differing only in the simply connected or statistically isotropic covariance benchmark. We denote this matched benchmark by $\mathcal{L}_{\mathrm{bench}}$. The first point at which topology can become a likelihood claim is

$$\Delta \ln \mathcal{L} = \ln \mathcal{L}_\Gamma - \ln \mathcal{L}_{\mathrm{bench}}. \tag{151}$$

### D. Transfer diagnostics

Blind-injection recovery, matched-circle rejection, and low-$\ell$ covariance comparisons are necessary implementation diagnostics for any chosen quotient. They are not part of the homogeneous classification theorem and are stated explicitly in Appendix A.

### E. Closure-route ledger

After the preceding definitions, the remaining solution space is finite in structure. In the current sector, pure pressureless behavior requires $F(n) = \mu c^2 n$ after any additive vacuum counterterm has been removed and assigned consistently to the vacuum sector; nonlinear stable $F$ defines a different EFT unless its pressure and sound speed are negligible. In the fixed-four-form sector considered here, the amplitude routes are continuous $q_b$, a single flux lattice, a dense multiflux lattice, a phenomenological constant, or no branch. In the topology sector, topology can enter only through

$$\mathrm{Vol}(\Sigma_\Gamma), \quad P_\Gamma, \quad m_\Gamma^{(r)}(n), \quad \Xi^\Gamma, \quad \mathrm{Cov}^\Gamma, \quad \pi_\Gamma(x_o). \tag{152}$$

It cannot determine $\Omega_c$, $\Omega_\Lambda$, or $\Lambda_{\mathrm{eff}}$ without $\mu_{\mathrm{w}}$, $e_F$, $A_0$, $H_0$, and branch data.

**Theorem 35** (Closure-route ledger)**.** *Within the declared FLRW residual, Brown-current, fixed-four-form, and quotient-covariance ledgers, every strict extension of the minimal homogeneous residual enters through at least one of the following routes:*

1. $\mathcal{R}_q \neq 0$ *or* $\mathcal{R}_{\mathrm{w}} \neq 0$*: a non-minimal homogeneous equation of state.*
2. $\delta p_{\mathrm{w,rf}} \neq 0$, $\Pi_{\mathrm{w}}^{\mathrm{GI}} \neq 0$, *or* $v_{s,\mathrm{w}}^2 \neq 0$*: non-dust perturbations.*
3. $\mathrm{d}N_F/\mathrm{d}\ln a \neq 0$, $\mathrm{d}e_F/\mathrm{d}\ln a \neq 0$, *or membrane transitions: a non-fixed vacuum branch.*
4. $\mathfrak{S}_{\mathrm{amp}}$ *uses non-topological normalization or branch data: amplitude selection.*
5. $\mathrm{Cov}^\Gamma$, $\pi_\Gamma(x_o)$, $M$, $N$, *and* $C_{\mathrm{fg}}$ *enter a likelihood: topology evidence.*

*Thus a manuscript-level claim is invariantly classified as a background closure claim, a perturbative source claim, a microscopic branch-selection claim, or a topology-likelihood claim.*

*Proof.* The atlas $(\mathcal{A}_q, \mathcal{A}_{\mathrm{w}})$ is algebraically invertible, so any homogeneous departure from the minimal residual is exactly a departure from the constant nonnegative submanifold, giving (i). The Brown-current representative is dust only on the linear branch $F(n) = \mu c^2 n$ after vacuum offsets are assigned to the vacuum sector, giving (ii). A four-form is a fixed vacuum source only on a fixed branch with no membrane transition, giving (iii). The topological decoupling theorem proves that amplitudes require non-topological normalization, giving (iv). Finally, the quotient projector becomes an observational topology claim only after it has been propagated to a masked, noisy data covariance and likelihood, giving (v). These cases exhaust only the objects introduced in the residual, source, topology, and likelihood ledgers. They do not claim to exhaust theories outside those ledgers,

such as modified gravity, different visible-sector subtractions, interacting dark sectors, non-perfect homogeneous residuals, anisotropic compact backreaction, non-FLRW backgrounds, nonlocal effective stress tensors, or quantum Casimir sources. □

## XIV. AUDIT TABLE AND BENCHMARK INTERFACE

The preceding laws are intended to be operational. A residual-closure audit can be implemented without assuming any particular microscopic dark-sector model. A minimal background interface is

$$\begin{aligned}(E(a), \epsilon_{\rm vis}(a), p_{\rm vis}(a), \Omega_K) &\longrightarrow (\epsilon_{\mathcal{D}}, p_{\mathcal{D}}) \\ &\longrightarrow (\mathcal{A}_q, \mathcal{A}_{\rm w}) \longrightarrow (\mathcal{R}_q, \mathcal{R}_{\rm w}, \mathfrak{C}_I).\end{aligned} \tag{153}$$

This interface is deliberately less ambitious than a Boltzmann solver. Its purpose is to decide whether the adopted residual is even eligible to be called minimal dust plus fixed vacuum before any topology, particle, or vacuum-origin claim is made.

This audit table is the proposed standard. It can be applied to a pure ΛCDM model, a $w_0 w_a$ dark-energy model, an interacting dark-sector model, a compact-topology model, or a modified-gravity reconstruction. The standard does not decide which model is true; it prevents the model from claiming more than its declared residual, microphysics, amplitude selector, and likelihood actually establish.

A practical program follows three separable papers or software modules. First, construct the residual and closure-defect module Eq. (153). Second, attach microscopic ledgers for current, vacuum, scalar-field, modified-gravity, or interacting sectors. Third, attach compact or noncompact transfer modules that propagate initial statistics into observable covariance and likelihood. Only the composition of all three modules is a predictive cosmological model.

## XV. CONCLUSION

The invariant content of the standard is the convention-declared residual atlas

$$(g_{\mu\nu}, T^{\rm vis}_{\mu\nu}, \text{Einstein convention}) \longmapsto T^{\mathcal{D}}_{\mu\nu} \longmapsto (\mathcal{A}_q, \mathcal{A}_{\rm w}), \tag{154}$$

with the affine subtraction law

$$T^{\rm vis}_{\mu\nu} \to T^{\rm vis}_{\mu\nu} + X_{\mu\nu}, \tag{155}$$

$$(\mathcal{A}_q, \mathcal{A}_{\rm w}) \to \big(\mathcal{A}_q + p_X,\, \mathcal{A}_{\rm w} - a^3(\epsilon_X + p_X)\big). \tag{156}$$

At the homogeneous locally-FLRW level, this atlas is lossless. The minimal residual is dust-like matter plus fixed vacuum if and only if

$$-p_{\mathcal{D}} = \text{constant} \geq 0, \qquad a^3(\epsilon_{\mathcal{D}} + p_{\mathcal{D}}) = \text{constant} \geq 0. \tag{157}$$

Equivalently,

$$\epsilon_{\mathcal{D}}(a) = \epsilon_{\rm w,0} a^{-3} + \epsilon_q, \qquad p_{\mathcal{D}}(a) = -\epsilon_q. \tag{158}$$

This proves exact local background degeneracy with cold dark matter plus Λ, provided the same amplitudes, curvature, visible subtraction, and perturbation covariance are used.

The natural minimal realization is a conserved Brown current with linear energy law plus a fixed four-form vacuum branch:

$$F(n) = \mu_{\rm w} c^2 n, \qquad T^q_{\mu\nu} = -\epsilon_q g_{\mu\nu}. \tag{159}$$

The Brown current closes the dust equation of state, not the dust abundance. The four-form closes the vacuum equation of state, not the vacuum amplitude. The compact causal ledger closes null-reach bookkeeping on the closed cover or a specified quotient, not the homogeneous stress tensor.

Thus the theory is invariantly admissible and background-closed at its stated level, but not predictively complete until amplitude selection and compact transfer likelihood are supplied. The status ledger is:

**Residual reconstruction::** closed.

**Homogeneous normal form::** closed.

**Dust equation of state::** closed by Brown current plus linear $F(n)$.

**Vacuum equation of state::** closed by fixed four-form branch.

**Dust amplitude::** not derived.

**Vacuum amplitude::** not derived.

**Local $R^3$ limit::** consistent.

**Compact causal ledger::** closed only after $A_0$, Γ, and $x_o$ are specified.

**Compact spectral closure::** requires quotient eigenmodes and projector.

**Perturbations::** source null tests stated; transfer closure incomplete.

**Observational likelihood::** not completed here.

The theorem also fixes the no-go boundary. Topology may restrict the global volume, causal reach, projected harmonic content, covariance matrix, and observer-position prior. It does not by itself provide the dimensionful current mass scale, four-form flux unit, branch rule, or likelihood evidence required to determine $\Omega_c$, $\Omega_\Lambda$, or $\Lambda_{\rm eff}$. Thus a compact quotient is a background, causal, spectral, and covariance input, not an abundance-origin mechanism.

The strongest form of the result is therefore a claim-discipline theorem: every FLRW dark-sector residual that

TABLE I. Invariant residual-closure audit. The table is a claim-control device: each row must be satisfied at its own level before a stronger interpretation is allowed.

| Audit question | Invariant object | Required statement |
|---|---|---|
| What is reconstructed? | $T^{\mathcal{D}}_{\mu\nu} = c^4 G_{\mu\nu}/(8\pi G) - T^{\rm vis}_{\mu\nu}$ | Metric, Einstein convention, and visible subtraction are declared. |
| Is the homogeneous residual dust plus vacuum? | $(\mathcal{A}_q, \mathcal{A}_{\rm w})$, $\mathfrak{C}_I$ | $\mathfrak{C}_I = 0$ and $\mathcal{A}_q, \mathcal{A}_{\rm w} \geq 0$ on the chosen interval. |
| Is dust microscopically realized? | Brown current $N^\mu$, energy law $F(n)$ | $\nabla_\mu N^\mu = 0$ and $F(n) = \mu_{\rm w} c^2 n$. |
| Is the dust amplitude selected? | $\mu_{\rm w}$, number rule, volume | A non-topological mass scale and branch rule, such as $N_J = \lvert Q\rvert$, are supplied. |
| Is vacuum equation of state realized? | Four-form branch $F_4$ | Fixed branch with $F_4 = f\varepsilon_4$ and no unsuppressed membrane jump. |
| Is vacuum amplitude selected? | $\widehat{q}$, $e_F$, $N_F$, branch prior | Continuous value, discrete flux unit, or landscape/prior is specified. |
| Is topology predictive? | $P_\Gamma$, $\mathrm{Spec}_\Gamma$, $m_\Gamma(n)$, $\mathrm{Cov}_\Gamma$ | Projector is propagated through transfer functions into observable covariance. |
| Is the likelihood comparison fair? | $\mathcal{L}_\Gamma - \mathcal{L}_{\rm bench}$ | The benchmark has the same local background, amplitudes, mask, noise, and foreground conventions. |

is exactly dust plus fixed vacuum is equivalent, at the invariant background level, to a conserved Brown current with linear energy law plus a constant four-form vacuum branch. Compact causal geometry supplies global causal and spectral closure ledgers, but not the amplitudes. A homogeneous equation-of-state departure is detected by $\mathcal{R}_q \neq 0$ or $\mathcal{R}_{\rm w} \neq 0$. A source departure is detected by nonzero rest-frame pressure, anisotropic stress, or sound speed. A vacuum-branch departure is detected by flux variation or membrane transitions. An amplitude claim requires non-topological normalization data. A topology claim requires a topology-covariance likelihood. These alternatives exhaust the declared causal residual compact-cosmology framework and make the manuscript self-contained at its stated level.

### Appendix A: Topology-likelihood diagnostics

For each candidate $\Gamma$, generate mock skies from $\mathrm{Cov}^\Gamma$ and matched controls from the benchmark covariance with the same local expansion history, curvature, visible sector, amplitudes, mask, beam, foreground model, and noise. Hide the injection labels and run both pipelines. With a threshold $\Delta_\star$, report

$$\mathrm{FPR} = P(\Delta \ln \mathcal{L} > \Delta_\star | \mathrm{bench}), \tag{A1}$$

$$\mathrm{FNR} = P(\Delta \ln \mathcal{L} < \Delta_\star | \Gamma), \tag{A2}$$

$$\mathrm{power} = 1 - \mathrm{FNR}. \tag{A3}$$

A topology claim is not likelihood-grade unless the pipeline distinguishes quotient covariance from the benchmark covariance on blind synthetic data before real-sky application.

For matched-circle and low-$\ell$ exclusions define

$$r_{\rm inj}(x_o) = \frac{1}{2} \min_{\gamma \neq e} d(x_o, \gamma x_o). \tag{A4}$$

If $r_{\rm inj}(x_o) \geq \chi_*$, the last-scattering sphere predicts no matched circles for that observer. If $r_{\rm inj}(x_o) < \chi_*$, each holonomy with $d(x_o, \gamma x_o) < 2\chi_*$ predicts a circle pair. In spherical geometry the circle radius obeys

$$\cos \alpha_\gamma = \frac{\tan[d(x_o, \gamma x_o)/2]}{\tan \chi_*}, \tag{A5}$$

with relative phase/twist fixed by the holonomy. Let $S^\Gamma_{\rm circles}$ denote the resulting maximized circle statistic, and let

$$\chi^2_{\text{low-}\ell,\Gamma} = d^\dagger_{\rm low} C^{-1}_{{\rm low},\Gamma} d_{\rm low} \tag{A6}$$

be the low-$\ell$ covariance statistic. The admissible quotient set is

$$\begin{aligned} \mathcal{A}_\Gamma = \{\, \Gamma, x_o, R, \eta : S^\Gamma_{\rm circles} < S_{\rm max}, \\ \chi^2_{\text{low-}\ell,\Gamma} < \chi^2_{\rm max}, \quad \mathcal{L}_\Gamma \text{ acceptable} \,\}. \end{aligned} \tag{A7}$$

If this set is empty, the chosen quotient implementation is excluded. If it is nonempty, the quotient is compact-compatible but not detected unless its evidence relative to the matched benchmark remains significant after blind injection calibration.

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